KICS | UET LAHORE | HUAWEI HCCDA

Kubernetes: Essential kubectl

One-page classroom reference | Namespace: kics-lab

01 Discover & Inspect

`kubectl get nodes -o wide`
Cluster node status and IPs

`kubectl get ns`
Show namespaces

`kubectl get pods -n kics-lab -o wide`
Pods, nodes, IP and readiness

`kubectl get deploy,rs,svc -n kics-lab`
Deployments, ReplicaSets, Services

`kubectl describe pod POD -n kics-lab`
Pod details, events, probe failures

02 Troubleshooting

`kubectl logs POD -n kics-lab`
Read container logs

`kubectl logs POD --previous -n kics-lab`
Logs from crashed previous container

`kubectl get events -n kics-lab`
Check recent warnings and failures

`kubectl top pods -n kics-lab`
CPU / RAM (Metrics Server required)

03 Safe Updates

`kubectl rollout status deploy/kics-blue -n kics-lab`
Wait for rolling deployment

`kubectl rollout history deploy/kics-blue -n kics-lab`
Inspect previous rollout revisions

`kubectl rollout undo deploy/kics-blue -n kics-lab`
Undo last Deployment revision

04 Manage Deployments

`kubectl scale deploy/kics-blue -n kics-lab --replicas=2`
Change desired replicas

`kubectl rollout restart deploy/kics-blue -n kics-lab`
Recreate pods using rolling update

`kubectl get hpa -n kics-lab`
Inspect Horizontal Pod Autoscaler

`kubectl get cm,secret -n kics-lab`
List ConfigMaps and Secret names

05 Services & Connectivity

`kubectl get svc -n kics-lab`
See ClusterIP and NodePort

`kubectl get endpoints -n kics-lab`
Service backend endpoints

`kubectl port-forward svc/NAME 8080:80 -n kics-lab`
Forward local port (keep terminal open)

`kubectl exec -it POD -n kics-lab -- sh`
Open shell (if image contains sh)

06 YAML & Context

`kubectl apply -f app.yaml`
Create/update resources declaratively

`kubectl diff -f app.yaml`
Preview desired-state changes

`kubectl delete -f app.yaml`
Delete resources in selected YAML

`kubectl config current-context`
Identify active cluster context

Using this lab ZIP? The ECS already has kubectl inside K3s.

`./kubectl.sh get pods -n kics-lab -o wide`
Run ./kubectl.sh followed by the remaining command; replace POD/NAME with real values.